College of Computing and Information Technology

Capstone Student Daily Activity Report

(only starting from the week when a project was allocated to your group)

Student Name	Ahmed Hedyed	**Student ID**	60106275
Supervisor Name	Dr. Adnan Adnan	**Mobile No. (optional)**	55563685
Degree Study Program Name	Bachelor of Science in Data and Cyber Security (Data Science and AI)	**Week No.**	3 and 5
Project Title	AI-Driven Fradulent Call Detection System		
Name of Group Members and Indicate the Leader	Mohammed Al-Sharshani (College of Computing & IT) Yousef Hadi (College of Computing & IT) Ahmed Hedyed (College of Computing & IT) Ahmed Radwan (College of Computing & IT) Ahmed Abdelrahim (College of Computing & IT) Shahd Al-Temimi (College of Computing & IT)		

Notes:

- You only need to report for days when you did work on the project. It is not necessary that you report on the project for every day of the week. Our expectation is a minimum engagement of 9 hours per week.
- We understand that you may work on the project over various start and end time periods per day, and thus, in such a case, please list all the relevant time periods for the day.
- You can report your project-related activities and tasks as detailed as you want. You can include figures, tables, snapshots, or simply anything that helps you capture the work that you did.
- For project tracking, a logbook is a very important document because, after several weeks, you may not remember the details of what you did. So, consider this document as your project's "memory bank", and it will serve as a reference when you do your regular group progress report and the eventual final report.
- The logbook must be done individually. Everyone will submit their own logbook, whereas the progress report will be a summary or synthesis of the group's progress till the submission week. Progress report is a different document than the logbook. A template for the progress report will be shared.
- Each logbook should be completed and submitted by the due date indicated on D2L Dropbox. This will be reviewed by the supervisor. Late submissions may not be scored.

DAY / DATE / TIME		ACTIVITIES AND TASKS PERFORMED (please include all relevant details)
Sunday	**1 – 2 – 2026**	• Conducted an internal team coordination meeting to formally initiate the Capstone implementation phase. • The team discussed the transition from the Practicum prototype into a fully developed application system. • A high-level system architecture was outlined, covering the Frontend, Backend, Model API integration, and Database components. Additionally, the dataset was reviewed to assess its quality, and several missing telecom metadata features were identified. **Individual Work:** • Reviewed and refined previously developed BERT training scripts. • Cleaned and organized model files. • Evaluated dataset limitations, particularly class imbalance and missing metadata. • Drafted an initial backend integration roadmap.
Start-End Time Period (s)	8:00 pm – 10:00 pm	
Monday	**2 – 2 – 2026**	• Initiated backend system planning using FastAPI. • The structure of API endpoints was discussed, including prediction, logging, and dashboard data endpoints. • A MongoDB schema was designed to store prediction results and call records efficiently. **Individual Work:** • Researched secure API authentication mechanisms. • Studied best practices for scalable backend architecture and deployment. • Documented the proposed backend architecture diagram.
Start-End Time Period (s)	8:00 pm – 10:00 pm	
Tuesday	**3 – 2 – 2026**	• Started frontend exploration and interface design discussions. • Identified essential UI components such as prediction output display, confidence score visualization, and fraud category labeling. Dashboard visualization requirements were also reviewed. **Individual Work:** • Designed a preliminary wireframe for the frontend layout. • Compared frontend development approaches (React framework vs simplified dashboard solution). • Researched effective fraud alert presentation models for user experience optimization
Start-End Time Period (s)	8:30 pm – 10:30 pm	
Wednesday	**4 – 2 – 2026**	**Weekly Supervisor Meeting – Dr. Adnan – 4:00 pm - 5:00pm** • Recorded detailed meeting notes. • Updated the project tracker and Gantt chart. • Drafted a structured dataset acquisition justification outline.
Start-End Time Period (s)	8:00 pm – 10:00 pm	
Thursday	**5 – 2 – 2026**	

Start-End Time Period (s)	9:00 pm – 10:00 pm	• Commenced the initial FastAPI backend setup and structured project directories (Model, API, Frontend, Database). Continued telecom dataset evaluation. **Individual Work:** • Prepared a formal dataset request draft addressed to Vodafone and Ooredoo. • Reviewed data privacy and regulatory compliance considerations. • Documented backend configuration procedures.

DAY / DATE / TIME		ACTIVITIES AND TASKS PERFORMED (please include all relevant details)
Sunday	**8 – 2 – 2026**	• Integrated the trained BERT model with a FastAPI test endpoint and initiated real-time prediction testing. Designed MongoDB logging structure for storing prediction outputs. **Individual Work:** • Measured model inference latency. • Optimized input preprocessing prior to API invocation. • Documented the integration workflow for reporting purposes.
Start-End Time Period (s)	8:00 pm – 10:00 pm	

Monday	**9– 2 – 2026**	• Continued frontend development and refined fraud classification display components. Evaluated potential dataset augmentation strategies.. **Individual Work:** • Researched telecom fraud patterns specific to the Qatar region. • Compared the TeleAntiFraud dataset with the current pooled dataset. • Updated dataset comparison documentation.
Start-End Time Period (s)	8:00 pm – 10:00 pm	

Tuesday	**10 – 2 – 2026**	• Conducted a structured dataset evaluation session, comparing public telecom datasets with the internally cleaned dataset. Discussed collaboration feasibility with telecom providers. • **Individual Work:** • Prepared a structured dataset comparison table. • Identified missing Call Detail Record (CDR) features required for realistic fraud simulation. • Drafted dataset enhancement recommendations.
Start-End Time Period (s)	8:30 pm – 10:30 pm	

Wednesday	**11 – 2 – 2026**	**Supervisor Meeting – Dr. Adnan – 4:00 pm - 5:00pm** • Demonstrated backend API progress. • Reviewed dataset acquisition efforts and discussed scalability considerations. • Supervisor advised continuing system implementation while pursuing dataset collaboration opportunities. • **Individual Work:** • Updated technical documentation based on supervisor feedback. • Refined backend improvement strategy. • Adjusted Gantt timeline for the upcoming implementation phase.
Start-End Time Period (s)	8:00 pm – 10:00 pm	

Thursday	**12 – 2 – 2026**	• Tested frontend-backend interaction and enhanced API response format to include fraud category, confidence score, and timestamp. Reviewed overall system stability. • **Individual Work:** • Updated technical documentation. • Cleaned and organized project repository structure.
Start-End Time Period (s)	9:00 pm – 10:00 pm	

		• Prepared weekly summary report for Capstone tracking